Rahul Kothagundla

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Professional Summary

Software Engineer with 1.5+ years designing scalable backend systems that handle real-time data at scale. At TCS, completely re-architected a critical data pipeline that was choking under load—cut processing time by 70% while keeping data integrity rock solid across 50,000+ daily transactions. Looking to bring this hands on experience to JPMorgan Chase's technology teams, where I can contribute to building the next generation of financial systems.

Technical Skills

Languages: Python, Java, C++, SQL, C
Backend & Distributed Systems: Apache Kafka, Apache Spark, FastAPI
Databases & Caching: PostgreSQL, Redis, SQL Server
Cloud & Infrastructure: Azure, AWS, Databricks, Docker, Docker Compose, CI/CD, Git
Core Strengths: System Design, Data Structures & Algorithms, Agile Development, Production Operations

Professional Experience

Tata Consultancy Services — June 2024 – Present
Systems Engineer — *Hyderabad, India*

- Inherited a data pipeline that was falling over during peak hours reports that should've taken minutes were stretching to 15+ minutes and business users were getting frustrated. Dug into the architecture, realized we were doing way too much computation in the reporting layer. Re-engineered the whole thing to push the heavy lifting into Databricks backend processing. Result? Cut processing time to 7 minutes (70% faster) and executives actually started using the reports in real-time for decisions.
- Built a monitoring system that watches our data pipelines 24/7 and automatically recovers from failures before anyone notices. The old system would silently fail and we'd only find out when someone complained about wrong numbers. Now it catches issues proactively, backs up the data and either fixes itself or pages me. We've handled 50,000+ records daily for months with zero data loss which matters a lot when you're dealing with financial data.
- Something I'm particularly proud of is creating a knowledge base with documentation and video walkthroughs that turned 7-week onboarding into 5 days. New engineers were spending weeks just trying to understand the system before they could contribute. Now they're productive almost immediately. Got a 5-star performance review, which felt good, but honestly the best part was seeing new teammates actually enjoying the onboarding instead of being overwhelmed.

Atomstate Technologies — June 2023 – September 2023
Machine Learning Engineer — *Hyderabad, India*

- Built an NLP sentiment analysis engine that could process thousands of social media posts and news articles to gauge public opinion. The whole pipeline was automated data comes in, model runs inference, results get stored. It was a deep dive into working with messy, unstructured text data at scale.

Zummit Infolabs — November 2022 – March 2023
Junior Data Scientist — *Remote*

- My first real job learned a ton about taking messy business data and turning it into something useful. Worked a lot with NLP to extract insights from unstructured text, building models that could parse through documents and pull out the important stuff automatically.

Technical Project

Real-Time Stock Market Data Streaming Platform | *Python, Kafka, Spark, PostgreSQL, Redis, Docker* — 2025

- Built this because I wanted to understand how real trading systems work, how do you process thousands of stock price updates per second without dropping data or showing stale information? Started with a simple idea: stream live market data and make it queryable in real-time. Ended up with a distributed system that handles 1,000+ events per second with sub-30 second latency. The fun part was making it fault-tolerant services can crash and restart and the system just keeps running. Hit 99.9% uptime during testing.
- The architecture uses Kafka (3 partitions for parallel processing), Spark Structured Streaming with exactly once semantics to prevent duplicate trades and PostgreSQL with composite indexes for fast time-series queries. Added Redis caching that cut database hits by 60% implementing proper cache invalidation was trickier than expected, but critical for preventing stale data. Containerized everything with Docker Compose, complete with health checks and restart policies for self healing. The system handles node failures, network blips and database connection drops without manual intervention, this kind of production ready resilience is exactly what matters for financial infrastructure.

Education

VNR Vignana Jyothi Institute of Engineering and Technology — Hyderabad, India
Bachelor of Technology in Computer Science & Engineering – Data Science (CGPA: 8.55/10) — *August 2020 – May 2024*

Certifications

AWS Cloud Quest: Generative AI Practitioner (Jan 2026)

Amazon ML Summer School 2023